

Temperature effects on the hydraulic properties of unsaturated rooted soils

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Abstract

Soil temperature of vegetated landfill earthen covers varies with time during their service, due to heat released by the degradation of municipal solid waste. In this study, an experimental campaign to capture soil water-retention curve (SWRC) and hydraulic conductivity function (HCF) of unsaturated rooted soils was conducted at different temperatures (25, 45, and 60 °C). Simplified evaporation tests were conducted by subjecting soil specimens to a prolonged drying period at maintained temperature. Based on the responses of matric suction and water content, SWRC and HCF were determined by the instantaneous profile method. It was found that increasing temperature reduced water-retention capacity, due to root shrinkage, thermal expansion of liquid phase, and reduction in forces of capillarity and adsorption. Soil hydraulic conductivity was enhanced by the increase in soil temperature due to reduction of water viscosity and potential preferential flow at root channels induced by root shrinkage. Temperature effects on SWRC and HCF of rooted soils are suction-dependent. A new nonisothermal SWRC model was developed to capture the dual effects of roots and temperature. By using the same set of calibrated parameters, the model demonstrates its ability to predict SWRCs reasonably well within the range of temperature investigated.

Key words: unsaturated soils, vegetation, temperature effects, soil water retention curve, hydraulic conductivity

1. Introduction

Landfilling is one of the most prevailing ways of waste treatment nowadays because of high feasibility, financially affordable cost, and relatively low environment risk. After waste layer being fully loaded, a final cover layer (commonly earthen cover) is required to separate waste layer from external environment. During the service life of landfill earthen cover, a main potential risk is hazardous leachate. If stormwater infiltrates into waste layer easily, the resulting leachate likely contaminates local aquatic environment, for instance, surface water body, groundwater, and potable water (Costa et al. 2019). An ideal landfill earthen cover is expected to be soils with high water-retention capability and low hydraulic conductivity to minimize waste leachate (Ni et al. 2020).

Nowadays, vegetation is widely utilized in the earthen cover, because it contributes to the removal of infiltrated rainwater from earthen cover to the atmosphere through evapotranspiration (Ng 2017; Ng et al. 2022). Effects of roots on hydraulic properties of soils at normal temperature depend on various factors: (a) dominant processes (e.g., root growth and decay), (b) root characteristics (e.g., root density, root diameter, root system type, etc.), and (c) soil properties (e.g., soil texture and degree of compaction). Root-induced changes of soil properties at normal temperature have been widely reported in previous studies as follows. (1)

Root water uptake and following transpiration increase soil suction and thus reduce moisture content of cover layer (Ng et al. 2014; Garg and Ng 2015). (2) Soil shear strength (especially soil cohesion) and soil aggregate stability are enhanced by roots (Jotisankasa and Taworn 2016; Leung et al. 2017; Mahannopkul and Jotisankasa 2019; Rajamanthri et al. 2021; Zhu et al. 2022). (3) Growing roots tend to lower soil gas permeability through filling soil pores (Chen et al. 2023). However, mature vegetation has been observed to be detrimental for reducing greenhouse gas emission in landfills due to increase of gas permeability (Ni and Ng 2019). (4) Soil hydraulic conductivity could be reduced due to actively growing roots, which occupy macropores and suppress cracks (Jotisankasa and Sirirattanachai 2017; Nguyen et al. 2020). (5) Hydraulic conductivity could be enhanced through preferential flow path formed by decaying or dead roots (Ghestem et al. 2011; Ni et al. 2019a, 2019b). Nevertheless, most previous studies on root–soil-interaction were conducted at normal temperature.

Decomposition (mainly anaerobic decomposition) of municipal solid wastes (MSW) in the waste layer produces greenhouse gas (e.g., methane), accompanied with heat energy (Feng et al. 2017; Chettri and Reddy 2021). The heat energy released would heat up the overlying cover layer, typically over 60 °C (Chettri and Reddy 2021). During decades of service of landfill covers, soil temperature would gradually decrease to

the ambient due to the reduction rate in the decomposition of MSW. Unlike the greenhouse gas, the released heat energy and its effects on performance of earthen cover are usually neglected. The temperature effects on other soil-based infrastructures have been widely reported. Temperature changes caused differentiated settlements in pavements, railway embankments, and airport runways (Zhou and Ng 2016b). Large temperature variations were related to potential failure areas in earth dams (Chen et al. 2018). Also, the stability of slope (especially silty and clayed slopes) was influenced by soil temperature (Thota and Vahedifard 2021) and matric suction (Ng and Shi 1998; Ng et al. 2003). As another soil-based infrastructure, vegetated earthen cover is also probably affected by soil temperature.

Single effect of root or temperature on soil hydraulic properties, such as soil water retention curve (SWRC) and hydraulic conductivity function (HCF), has been widely investigated. Many models have been proposed to determine temperature-dependent SWRC (Grant and Salehzadeh 1996; She and Sleep 1998; Zhou and Ng 2016a; Leung et al. 2020), root-dependent SWRC (Ng et al. 2016; Ni et al. 2019a), and biopolymer-dependent SWRC (Zhou et al. 2020). An HCF model could be further developed from a SWRC model using the approach by Mualem (1976). However, there is still a paucity of research on the development of nonisothermal hydraulic models of rooted soils.

The objective of this study is to investigate the effects of temperature on the SWRC and HCF of unsaturated rooted soils. The first part of this study was to design and conduct a comprehensive experimental campaign to provide experimental data to characterise any temperature dependence on these two hydraulic properties of rooted soils. The second part was to derive and propose a new SWRC model that considers the dual effects of roots and temperature. The data obtained from the first part were used to evaluate the predictability of the new model from the second part.

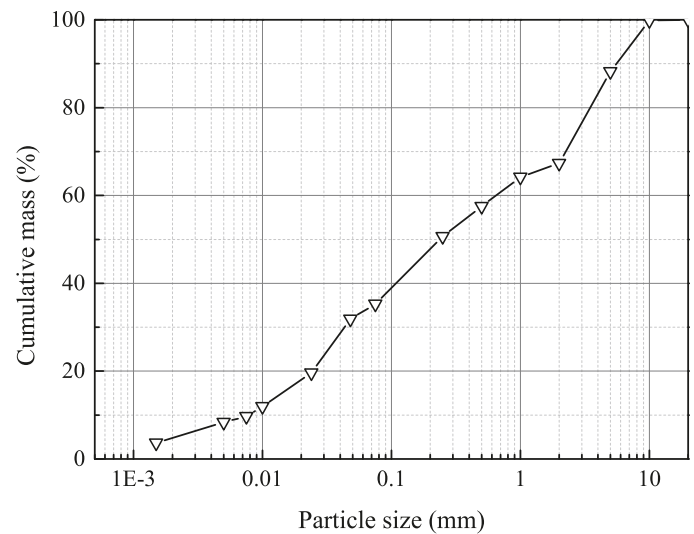
2. Materials and test programme

2.1. Plant cultivation and specimen preparation

In this study, a field site (5 × 5 m, 0.3 m depth) in Shenzhen, China (114°03'E, 22°32'N), was built for plant cultivation. Fourteen tons of local granitic residual soils were crushed, sieved to remove large particles (diameter ≥ 20 mm), and air-dried. Then, the soils were mixed with water to the targeted water content of 20% (w/w) and sealed by plastic cloth for 24 h to achieve moisture equilibrium. The moist soils were then compacted in five layers at the site. The surface of each layer was scratched to ensure intimate contact between successive layers. The particle-size distribution of the test soil is shown in Fig. 1. The soil was poorly graded and contained more than 85% coarse content (sand + gravel). Based on the standard proctor test, the maximum dry density of the soils was 1.64 g/cm³, the corresponding optimum water content is 18.5% (w/w). No fertilizer was added in soils throughout the experiment.

Vetiveria zizanioides (known as vetiver grass) was tested in this study. This species is commonly found in tropical to

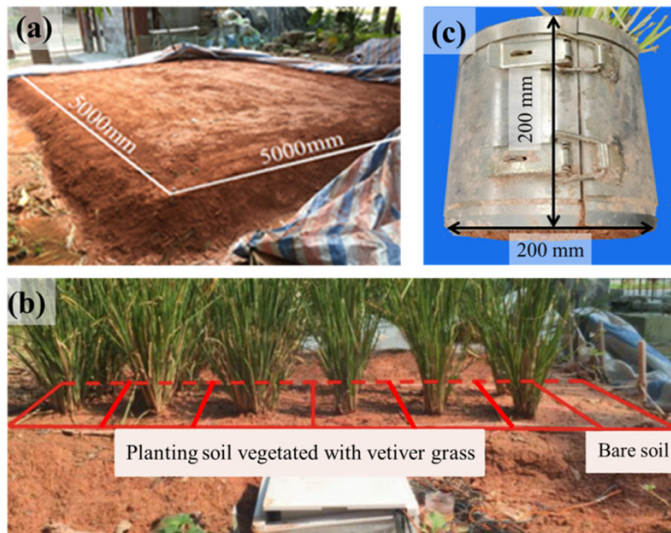
Fig. 1. Particle-size distribution of test soil.



subtropical regions (e.g., India and Malaysia), and it is tolerant to hot and humid environments in these regions. This species has been widely used to enhance the ecological values and biodiversity of earthen infrastructure such as slopes and landfill covers, because it is inexpensive, has high survival rate (i.e., low maintenance costs), and has impressive deep-rooted trait and biomechanical properties, which is beneficial to erosion and stability controls (Nguyen et al. 2019; Wu et al. 2021). The seed-born vetiver used in this study is native to southern China (Liu et al. 2021). Grasses were bought from a nursery of a local supplier. Then, they were transplanted to the site with a planting density of 100 plants/m². They were cultivated for seven months, during which the average monthly temperature and air relative humidity were 23.70 ± 4.85 °C (mean ± standard deviation) and 77 ± 6%, respectively. The site received less than 100 mm rainfall per month in the first 5 months, while the remaining 2 months had a rainfall amount of 200–300 and 400–500 mm, respectively. After cultivation, soil columns with 0.2 m tall and 0.2 m diameter were sampled from the top 0.2 m of the field soil for laboratory testing (as shown in Fig. 2c). After field sampling, the top and the bottom of the soil columns were covered by a plastic film to minimise water loss. To minimise any effect of root decay, (1) the aboveground biomass was not completely removed during storage, and (2) all the experiments (see the test programme below) were conducted within six days after the field sampling.

In this study, the rooted soil specimens were tested at 3 temperatures (25, 45, and 60 °C, denoted as VS-25, VS-45, and VS-60, respectively). Three replicates were measured at each targeted temperature. The soil temperature of a landfill cover primarily depends on the stage of service life because of changes in decomposition of MSW (mainly anaerobic decomposition with methane generation). In the early stage, heat released from the MSW due to decomposition processes is substantial, resulting in relatively high soil temperature within the cover (typically over 60 °C; Chettri and Reddy 2021). Following decades of service, the reduction of heat release

Fig. 2. Plant cultivation and test specimen sampling: (a) site building for plant cultivation, (b) grass growth, and (c) soil sampling.



causes a drop in soil temperature to approximately 45 °C and eventually to the ambient (about 25 °C). Therefore, the behaviour of VS-25, VS-45, and VS-60 measured from this test programme represents the soil temperature conditions at the early, middle, and final stages of the service life of a landfill cover, respectively. Some root traits of VS-25, VS-45, and VS-60 obtained are summarised in Table 1. Small differences (no more than 7%) exist among them regarding both the total root biomass and root distribution.

2.2. Measurements of soil hydraulic properties

Hydraulic properties of unsaturated rooted soils, including SWRC and HCF ($K(\psi)$), at three targeted temperatures were measured by the simplified evaporation method (Schindler and Müller 2006). Figure 3 shows the schematic of the test set-up. The set-up includes a temperature-controlled container, which was thermally insulated by plastic foam, insulation coating, and thermal insulation tape. Each soil sample, after saturation with water, was placed in a flexible wall (i.e., a latex film) to minimize sidewall evaporation. The air in the container was then heated by electric heating wires to maintain the targeted air temperature (i.e., 25, 45, or 60 °C), and the respective soil samples were left dried within the container. To measure changes in soil matric suction during the evaporation process, two high-capacity tensiometers (HCTs), which were calibrated against gauge pressure (Chen et al. 2015), were installed and attached to the soil specimens. One HCT was installed at 25 mm from the top and the other was at 25 mm from the bottom of each soil specimen (200 mm tall) so that they were symmetric with respect to the central plane of the specimen. Thus, soil matric suction at the central plane could be determined by linearly interpolating the measurements made by these two HCTs. Trial tests showed that exposing the HCTs at a high-temperature environment (60 °C) for long test duration always resulted in cavitation in the HCTs.

Thus, heat dissipation suction sensors (HDSs) were used in the experiments conducted at 60 °C instead of the HCTs. HCTs and HDSs were both carefully calibrated, expecting compatible measurements similar to observations by Mendes and Gallipoli (2020). A load cell (model: ZNBS-P), with a working range of temperature from –20 to 70 °C, was used to measure the mass change of each specimen upon evaporation. A wireless data logging system, which was composed of an analogue to digital converter (ADC; model UL10AI, Liankong Intelligent Technology Co., Ltd., China), a power supply, transmitters, and receivers of wireless signal (DTK Electronics Inc., China), was adopted. The working temperature ranges of the ADC, power supply, and the signal transmitters were 10–90, 10–70, and –40–85 °C, respectively. All the data were collected and transmitted at a regular interval of 30 s. All experiments were completed within 60 h. Thus, any changes in the distribution and biomass of roots upon growth/decay were deemed insignificant to affect the test results.

Based on the mass of evaporated water and known saturated water content (θ_s), gravimetric water content at any time interval can be estimated. Hence, a drying SWRC can be constructed by relating the water content with matric suction. On the other hand, HCF at a given soil matric suction [$K_u(\bar{\psi})$] can be obtained by the Darcy's law (Schindler and Müller 2006), as expressed by

$$(1) \quad K_u(\bar{\psi}) = \frac{\delta V}{2A\bar{i}\delta t}$$

where δV is the volume of water evaporated during a time interval δt . $\bar{\psi}$ is the mean value of matric suctions measured by the two tensiometers, A is the cross-sectional area of the specimen, and $\bar{i}$ is the mean hydraulic gradient during δt , which could be expressed by considering the central difference, as follows:

$$(2) \quad \bar{i} = \frac{1}{2} \left(\frac{\psi_{i,\text{up}} - \psi_{i,\text{low}}}{\delta h} + \frac{\psi_{i+1,\text{up}} - \psi_{i+1,\text{low}}}{\delta h} \right) - 1$$

where δh is the height difference between the two HCTs or HDSs. At t_i , matric suctions measured by the lower tensiometer and the upper tensiometer were denoted by $\psi_{i,\text{low}}$ and $\psi_{i,\text{up}}$, respectively. At t_{i+1} , suctions measured by the lower tensiometer and the upper tensiometer were denoted by $\psi_{i+1,\text{low}}$ and $\psi_{i+1,\text{up}}$, respectively.

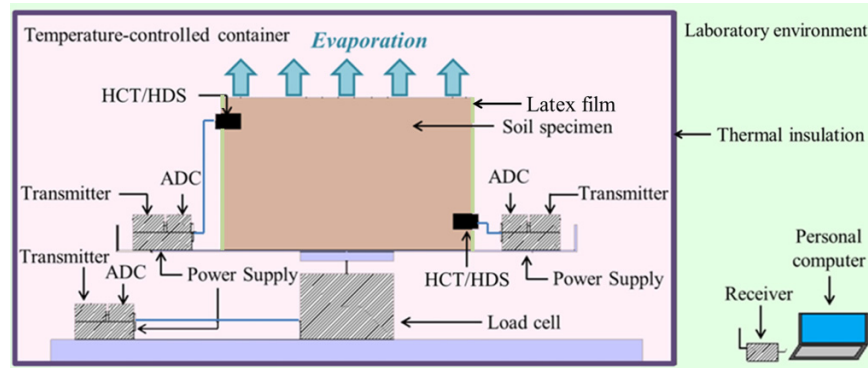
3. Interpretations of measured data

3.1. Measured soil water retention curves of rooted soils at different temperatures

Measured data of matric suction is related to volumetric water content and degree of saturation (S_r) in Figs. 4a and 4b, respectively. Note that S_r was calculated based on the measured gravimetric water content during the experiments and the known initial dry density of the soil samples. Physically, SWRC is a measure of the soil water-storage capacity for a given suction, which is influenced by volume change (Ng and Pang 2000; Chiu and Ng 2003). The soil type tested contained large coarse content (>85%) and thus had rather low plastic-

Table 1. Root traits of three types of soil specimens.

Specimen I.D.	Root volume density (R_v ; cm^3/cm^3)	Root length density (R_l ; cm/cm^3)	Root volume percentage in different depths (%)			
			0~5 cm	5~10 cm	10~15 cm	15~20 cm
VS-25	1.85×10^{-3}	16.67	72.5	17.0	10.0	0.5
VS-45	1.74×10^{-3}	13.23	70.0	18.0	10.5	1.5
VS-60	1.84×10^{-3}	15.45	69.0	17.5	12.5	1.0

Fig. 3. A schematic diagram showing the test set-up of the simplified evaporation tests at elevated temperatures.

ity. Indeed, there was no observable change in soil volume after drying. It is thus reasonable to assume that the soil was rigid during the drying process. The increase in temperature reduced the air-entry values (AEVs) and increased desorption rates. These observations imply that soils subjected to higher temperature are easier to be desaturated upon an increase in matric suction. The temperature-induced reduction of water-retention capability has similarly been observed for soils without vegetation (e.g., Salager et al. 2011; Leung et al. 2020). The temperature dependence is because of (a) the reduction in microscopic forces responsible for water retention (mainly capillarity and adsorption) and (b) the thermal expansion of the liquid phase (Salager et al. 2011). Similar explanation may apply to the case of vegetated soil in the present study.

In addition to the two thermal effects aforementioned that may generally exist in bare soils, the presence of roots in soil pores could affect the water-retention capability. Previous experimental evidences (e.g., Leung et al. 2015) and modelling efforts (e.g., Ng et al. 2016; Ni et al. 2019a) have already demonstrated that roots could either increase or decrease the soil water retention, depending on how the soil pore space was being physically altered by the changes in the root volume. When subjected to a higher temperature, the roots in the soil might be dehydrated and shrunk. The reduction in root dimensions (both diameter and length) means a reduction in the volume of root occupation in the soil pore space, causing an increase of soil void ratio (when treated root as a solid phase). According to the ink's bottle effect (Ng et al. 2009), the apparent "loosening" effect to the soil introduced by the root shrinkage upon high temperature may be a reason that contributed to the reduction of the soil water-retention capability. Interestingly, however, the effects of root shrinkage on the SWRC became less prominent

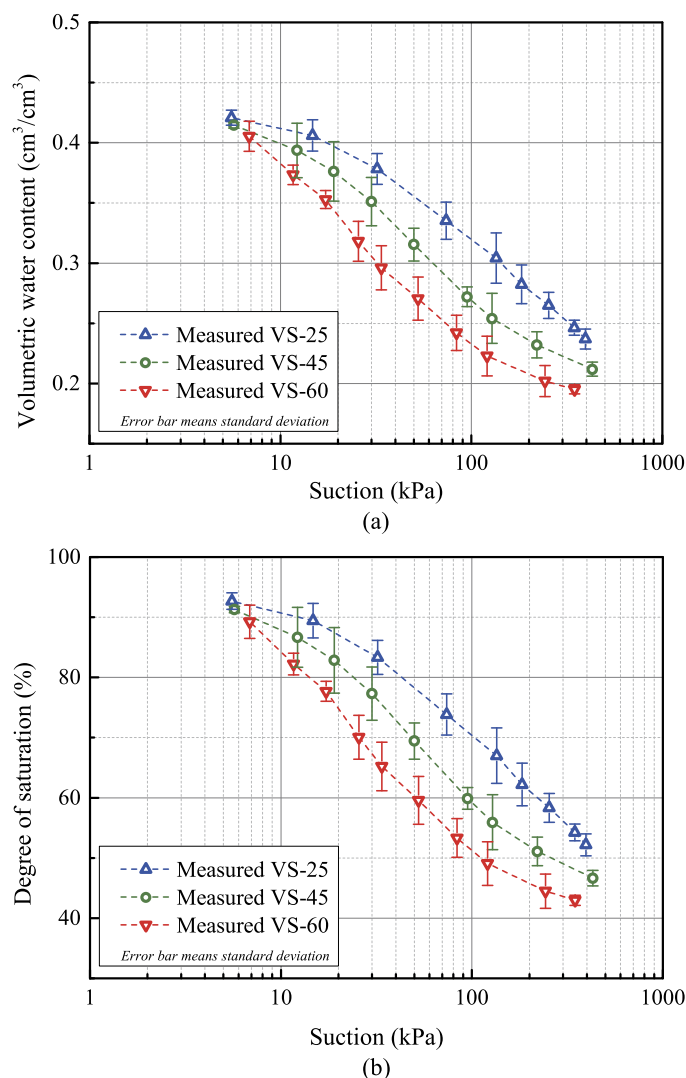
at high matric suction. Indeed, the values of S_r among the three cases seem to be converging at higher matric suction. At 350 kPa suction, the difference of S_r between VS-25 and VS-60 was approximately 11%, which was much smaller compared to the difference found in the intermediate suction (e.g., about 20% at 100 kPa). This may be somewhat expected because at high suction (i.e., relatively dry condition), the amount of root moisture losses in the three types of soil specimens were comparable, irrespective of the temperature experienced. Indeed, this kind of convergence did not exist for bare soils even at very high matric suction. For instance, the difference of S_r of clayed soils tested at 21 and 80 °C remained high (i.e., >20%) at 10 MPa (Salager et al. 2011).

From a practical point of view, soil specimens subjected to 60, 45, and 25 °C may represent the field conditions at early, middle, and final stages of the service life of an earthen cover, respectively. The variation of the water-retention capability shown from the measured SWRCs implies that temperature effects should be taken into account almost throughout the service life of the cover system to more correctly assess the water transport and balance calculation within the earthen covers. The fact that soil has reduced water-retention capability at higher temperature also means that the design of water and gas transport at the early stage of the service life is much more critical than that at other two stages, due to underestimation of leachate volume for the case that has otherwise ignored the temperature effect in the analysis.

3.2. Measured hydraulic conductivity function of rooted soils at different temperatures

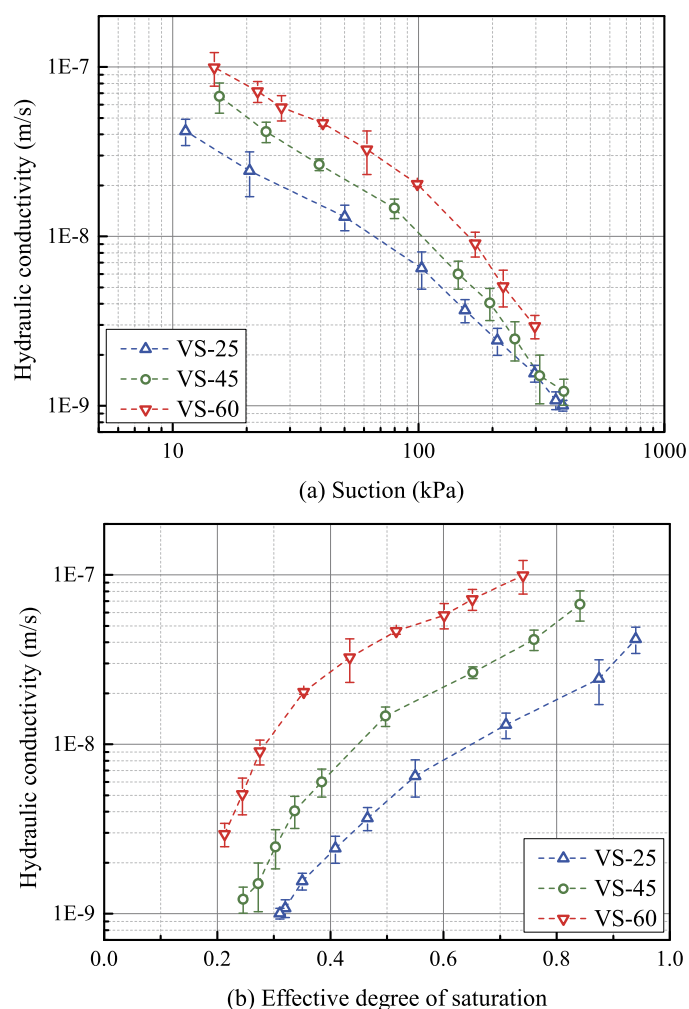
Figure 5 shows the variations of hydraulic conductivity of rooted soils, expressed in terms of either matric suction (Fig. 5a) or S_r (Fig. 5b). Evidently, in either case, the hydraulic conductivity at any suction or S_r increased with an increase in

Fig. 4. Measured soil water-retention curves of rooted soils at different temperatures: (a) volumetric water content versus suction and (b) degree of saturation versus suction. Each data point is expressed as mean $\pm$ standard deviation ($n = 3$).



temperature. Temperature-induced increase in the hydraulic conductivity has been commonly observed in bare soils, e.g., compacted bentonite (Ye et al. 2012), silty clay (Ren et al. 2014), clay (Ma et al. 2016), and standard HST95 Congleton silica sand (Leung et al. 2020). This is primarily because of the reduction of water viscosity at elevated temperature (e.g., water viscosity at 60 °C was about half of that at 25 °C). This mechanism also applies to the vegetated soils tested in the present study. In addition to the tortuous passages (i.e., connected soil pores) in bulk soils, which also exist in the case of bare soil, the presence of roots in the soil pore space may introduce additional water transport channels along the interfaces between roots and soils. As roots grow, a small zone (in mm scale) called rhizosphere soil would likely be formed in the vicinity of the roots, and this zone has active biochemical reactions and has rich contents of soil organic matters (SOMs). The SOMs in the rhizosphere act as a binding agent between the roots and soils. Soils subjected to elevated tem-

Fig. 5. Measured hydraulic conductivity function of rooted soils at different temperatures when relating with (a) matric suction and (b) degree of saturation. Each data point is expressed as mean $\pm$ standard deviation ($n = 3$).



perature would accelerate the decomposition of SOMs, causing a reduction of the binding effects (Zhu and Cheng 2011), promoting more water flow channels along the root-soil interface and thus causing an increase in the hydraulic conductivity.

Furthermore, in previous studies, it has been found that temperature effects on the hydraulic conductivity of bare soils are suction-dependent (Ye et al. 2012; Zhang et al. 2013). According to Ye et al. (2012), temperature-induced changes in water viscosity played a predominant role in high suction range, while temperature-induced changes in soil structure and water viscosity were both crucial when considering relatively low suction range. Based on the results presented by Zhang et al. (2013), the temperature effect on unsaturated hydraulic conductivity was limited (less than 10%) for a rather dry soil with water content less than 20% (by mass). Consistently, the test data shown in Fig. 5 demonstrate that the temperature effects on the hydraulic conductivity of rooted soils are also suction-dependent. The difference between VS-25 and VS-60 is more remarkable at relatively low

suction range (<150 kPa) as compared to suction range over 150 kPa. Furthermore, HCF of VS-45 is similar to that of VS-60 in low suction range whereas analogous to that of VS-25 at 300 kPa (as shown in Fig. 5a). This suggests that changes in soil HCF induced by sub-high temperature are important at near-saturation condition but insignificant at dry condition. Suction dependence of temperature effect for bare soils may be explained as follows. Elevated temperature reduces water viscosity and thus increases hydraulic conductivity, which is not significant in dry case when less volume of water was available in the soil pore space. For case of rooted soils, roots would shrink at either high temperature or low S_r , effectively creating preferential flow paths to facilitate the increase in hydraulic conductivity. If roots had shrunk due to high-temperature stress, they would be less vulnerable to latter drought stress. In practice, most leaching accidents occur in saturated earthen cover, since saturated hydraulic conductivity is much greater than unsaturated ones. Because elevated temperature significantly increases saturated hydraulic conductivity, leaching risk would be underestimated if temperature effects are ignored in the water balance calculation.

4. Model formulations

4.1. Modelling nonisothermal water-retention curve of rooted soils

A nonisothermal SWRC model of rooted soil was proposed, based on the following assumptions: (1) root effects on SWRC are exclusively because of the physical occupancy of roots in soil pores, (2) root volume changes (i.e., root expansion and shrinkage due to suction and temperature changes) are ignored, and (3) any effects of roots and temperature are uncoupled. To capture the temperature effect, the modelling approach initially proposed by Grant and Salehzadeh (1996) and later modified by others (e.g., She and Sleep 1998; Leung et al. 2020) was adopted, as follows:

$$(3) \quad S_e = \left\{ 1 + \left[\frac{\psi (T_r - \beta)}{\alpha (T_p - \beta)} \right]^n \right\}^{-m}$$

where S_e is the effective degree of saturation and expressed as eq. 4. ψ is the soil matric suction (kPa), β (unit: K) is an empirical material constant, and α (kPa) is a fitting parameter related to the AEV of the soil. T_r is the reference temperature [i.e., 298 K (25 °C) in this study]; T_p is the temperature under consideration. m and n are two dimensionless fitting coefficients and related by eq. 5:

$$(4) \quad S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}$$

$$(5) \quad m = 1 - \frac{1}{n}$$

The values of β varied from 346.4 to 448.6 K, primarily depending on the soil types and the wetting/drying process (i.e., imbibition or drainage) (Grant and Salehzadeh 1996; She and Sleep 1998; Leung et al. 2020). Note that all these existing studies calibrated the β based on the test data mea-

sured from bare soils. Furthermore, there is a limitation for the model, i.e., prediction range is limited by parameter β . If $T_r < \beta < T_p$, there exists non-integer power of a negative number (i.e., $\left[\frac{\psi (T_r - \beta)}{\alpha (T_p - \beta)} \right]^n$) during calculation. In most cases, non-integer power of a negative number is not calculable in the range of real numbers.

Next, the model proposed by Ng et al. (2016) may be used to introduce root effect into the isothermal SWRC of rooted soils, as follows:

$$(6) \quad S_r = \left[1 + \left(\frac{\psi e^{m_4}}{\alpha} \right)^n \right]^{-m}$$

To be consistent with the mathematical form of eq. 3, eq. 6 is re-expressed as eq. 7. The difference between S_e and S_r is minor for coarse soils, especially in low suction range:

$$(7) \quad S_e = \left[1 + \left(\frac{\psi e^{m_4}}{\alpha} \right)^n \right]^{-m}$$

where α , m , and n are the same as those used in eq. 3, and m_4 is a dimensionless fitting coefficient associated with root characteristics. Considering that effects of void ratio on SWRC of coarse-grained soils are negligible at high suction range, parameters m_4 , m , and n are related by eq. 8, according to Gallipoli et al. (2003) and Ng et al. (2016):

$$(8) \quad m \cdot n \cdot m_4 = 1$$

Hypothesising that the effects of roots on SWRC are exclusively because of the physical occupancy of roots in the pore space and considering the phase diagram of an unsaturated rooted soil, the void ratio of the rooted soil (e) may be expressed by eq. 9 (Ng et al. 2016):

$$(9) \quad e = \frac{e_0 - R_v (1 + e_0)}{1 + R_v (1 + e_0)}$$

where e_0 is the void ratio of the bare soil, and R_v is the root volume ratio (cm³/cm³), defined as the total volume of roots to the soil volume. In addition, as root volume changes were ignored in this study, the parameter R_v was considered to be a constant for each specimen.

Any coupled effect of roots and temperature on soil hydraulic properties was ignored in this study. By integrating eqs. 3 and 7, a new model for nonisothermal SWRC of rooted soils, namely RT-SWRC model, is expressed, as follows:

$$(10) \quad S_e = \left\{ 1 + \left[\frac{e^{m_4} (T_r - \beta)}{\alpha (T_p - \beta)} \psi \right]^n \right\}^{-m}$$

By including eqs. 5 and 8, eq. 10 becomes

$$(11) \quad S_e = \left\{ 1 + \left[\frac{e^{1/(n-1)} (T_r - \beta)}{\alpha (T_p - \beta)} \psi \right]^n \right\}^{-(1-1/n)}$$

4.2. Modelling nonisothermal hydraulic conductivity function of rooted soils

To capture the variations of hydraulic conductivity with matric suction, Mualem (1976) proposed the following equation by assuming spatially random distribution of soil pores

and then calculating the probability of pores encountering other pores:

$$(12) \quad K_r = S_e^\tau \left[\frac{\int_0^{S_e} \frac{1}{\psi(x)} dx}{\int_0^1 \frac{1}{\psi(x)} dx} \right]^2$$

where τ is a dimensionless fitting parameter, which is commonly assumed to be equal to 0.5 (e.g., [Mualem 1976](#); [Ni et al.](#)

[2019b](#)); K_r is the relative hydraulic conductivity and expressed as follows:

$$(13) \quad K_r = \frac{K_u}{K_s}$$

where K_s is the saturated hydraulic conductivity (m/s), and K_u is the unsaturated hydraulic conductivity (m/s).

Substituting the RT-SWRC model ([eq. 11](#)) into the Mualem model ([eq. 12](#)) results in a new RT-HCF model, as follows:

$$(14) \quad K_r(\psi) = \left\{ 1 + \left[\frac{e^{1/(n-1)} (T_r - \beta)}{\alpha (T_p - \beta)} \psi \right]^n \right\}^{-\frac{n-1}{2n}} \left\{ 1 - \left\{ 1 - \left\{ 1 + \left[\frac{e^{1/(n-1)} (T_r - \beta)}{\alpha (T_p - \beta)} \psi \right]^n \right\}^{-1} \right\}^{\frac{n-1}{n}} \right\}^2$$

5. Model performance

5.1. Validation of the new nonisothermal SWRC model of rooted soils

Based on the measurement of unsaturated soils vegetated with vetiver grass, SWRCs of rooted soils at 25, 45, and 60 °C are calibrated by the new RT-SWRC model ([eq. 11](#)). The parameter β is known to be soil-specific, so instead of directly adopting the values reported in the literature, this parameter was calibrated for the specific type of rooted soils examined in this study. Before calibrating β , the SWRC measured at 25 °C was used to obtain the parameters α and n using multi-nonlinear regression. Note that the reference temperature T_r was set to be 25 °C, so the parameter β does not play a role in the SWRC measured at this temperature. The calibration was conducted by varying the value of β until the predicted curves displayed the best fit with the measured ones, evaluated by the goodness of fit (R^2). It is found that it is possible to use a unique β of 355 K to predict the SWRCs of the rooted soils at three different temperatures ([Fig. 6](#)). The value of β falls within the typical range (346.4–448.6 K) reported in previous studies ([Grant and Salehzadeh 1996](#); [She and Sleep 1998](#); [Leung et al. 2020](#)). Interestingly, it can be observed from [Fig. 6a](#) that for the specific soil and roots tested, β was insensitive to the SWRC at 45 °C, but at a higher temperature of 60 °C ([Fig. 6b](#)), a careful choice of β is crucial in the prediction of SWRC since it controls the shape of the curve (or the desorption rate). All the parameters used for the RT-SWRC model, including β , are given in [Table 2](#).

For different soil specimens, identical e_0 , α , n , and β but different R_v were used (refer to [Table 1](#)) in RT-SWRC model. The performance of RT-SWRC model is given in [Fig. 7](#). It is found that the new model could effectively capture the SWRC measured at 45 °C for the entire range of suction. Also, the model worked particularly well in the suction range between 10 and 300 kPa for the specimens VS-25 and VS-60. As assumed, any coupling between the root and temperature effects on SWRC was not considered in the RT-SWRC model. Indeed, roots might shrink at high temperature as they dehydrate; thus, the R_v may be a function of temperature. Future work is needed to characterise such coupling effect to

Fig. 6. Calibration of the parameter β used in the RT-SWRC model: (a) VS-45 and (b) VS-60.

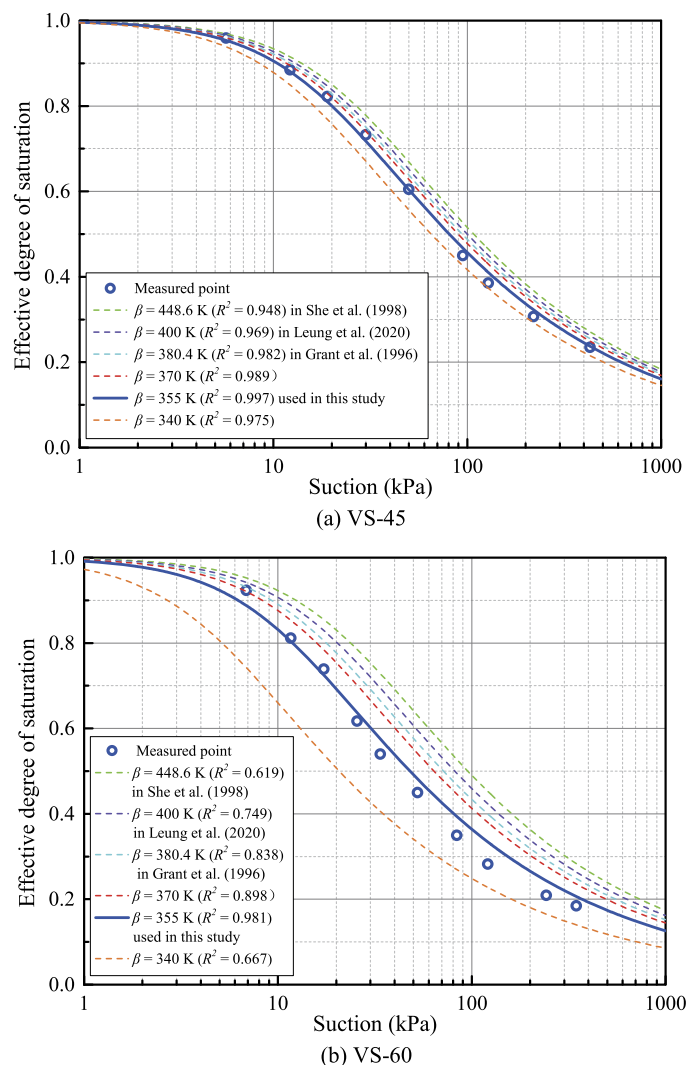
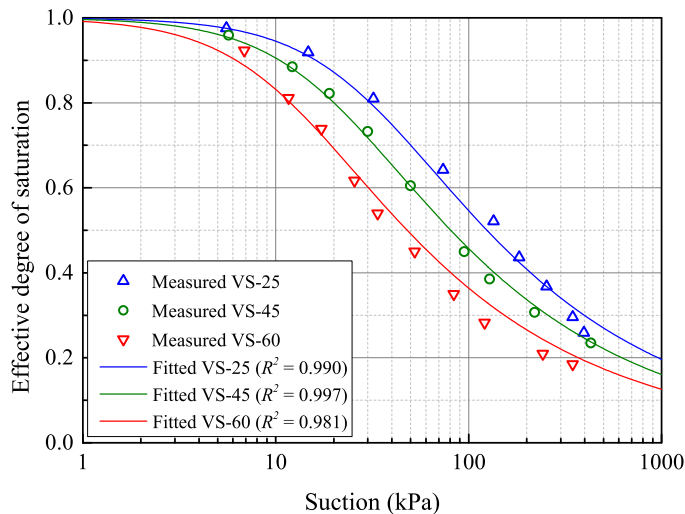


Table 2. Parameters used in new RT-SWRC model.

α (kPa)	n	e_0	R_v (cm ³ /cm ³)	β (K)	R^2
19.8	1.466	0.832	Refer to Table 1	355	>0.98

Fig. 7. Performance of new RT-SWTC model, considering dual effects of roots and temperature.

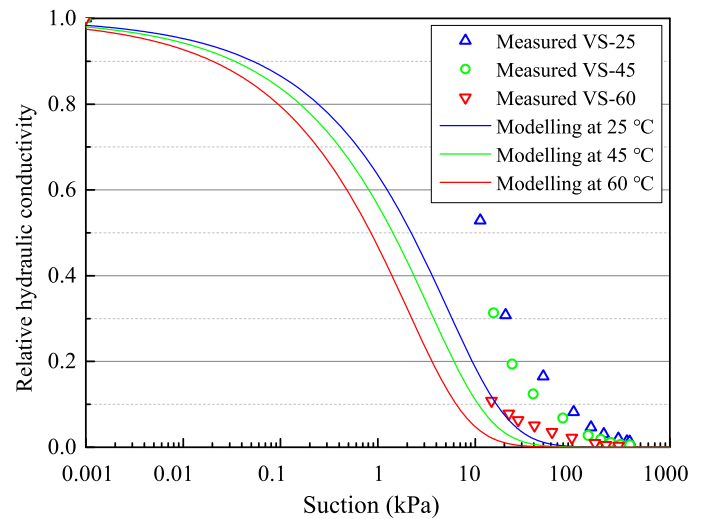


further improve the formulation of the model. Despite the satisfactory performance of the RT-SWRC model against the data obtained from vetiver grass, the universality of the proposed model to other species that have different root distributions needs to be examined, when more data of temperature-dependent SWRC of rooted soils are available.

5.2. Performance of the HCF model based on RT-SWRC

In this study, the HCF model based on RT-SWRC using the Mualem's approach was used to predict the relative hydraulic conductivity function of rooted soils. Consistently, both the measurements and predictions showed that at any given suction, qualitatively, the relative HCFs were lower when the temperature that the rooted soils experienced was higher (Fig. 8). However, quantitatively, the predictions of all three HCFs were far from the respective measurements. The large discrepancies are not surprising because the original formulation of Mualem (1976) does not take into account both the effects of roots and temperature. In the model, the distribution of soil pores is assumed to be spatially completely random, and the soil HCF is calculated based on the probability of pores' connection. Previous studies (e.g., Mualem 1976; Leung et al. 2020) have demonstrated excellent performance of the model to certain types of bare soils. For rooted soils, however, the soil pore size distribution is influenced by the presence of roots and the pattern or network that they distributed spatially in the soil (i.e., not complete spatial randomness). Because of root growth and water absorption, roots might expand and thus cause re-arrangement of surrounding soil particles (Lu et al. 2020). On the other hand, when roots

Fig. 8. Measured and predicted HCFs of rooted soils at three different temperatures.



shrink due to decay or when the plant is subjected to drought stress, some pore space previously occupied by roots might be "returned", creating preferential flow path and thus enhancing the soil hydraulic conductivity (Ghestem et al. 2011; Ni et al. 2019a). Under the effects of temperature, the soil structure might be changed (Zhou and Ng 2015; Chen et al. 2018; Thota and Vahedifard 2021), which further alters the root-soil interaction and the pore size distribution. Unfortunately, data about how roots alter the pore size distribution under different conditions (upon growing, decaying, etc.) and how this interaction is affected by temperature change are not available. Any improvement of the prediction HCF can be done when more research in this subject is conducted in the future.

6. Conclusions

This study aims to explore influences of temperature on the hydraulic properties, including SWRC and HCF of unsaturated soils vegetated with vetiver grass. An experimental campaign was carried out to measure these hydraulic properties at different controlled temperatures (25, 45, and 60 °C) in the laboratory. A new nonisothermal model was developed to capture the dual effects of roots and temperature on the SWRC of unsaturated rooted soils. Based on the measurements and modelling, the following conclusions may be drawn:

1. Increasing the soil temperature substantially reduced the soil water-retention capability for the suction range investigated. Specifically, the increase in soil temperature causes reduction in air-entry values and increase in desorption rate. This is due to (a) the reduction in microscopic forces responsible for water retention (e.g., capillarity and adsorption), (b) thermal expansion of the liquid phase, and (c) temperature-induced root shrinkages. Temperature effects on SWRC of rooted soils are significant in the intermediate suction (about 100 kPa).

2. Hydraulic conductivity of unsaturated rooted soil at any matric suction or S_r was enhanced by the increase in soil temperature. This is because of (a) the reduction of water viscosity coefficient and (b) potential preferential flow at the root channels due to root shrinkage. Temperature effects on HCF were more significant at near-saturation condition, as compared to dry condition.
3. A new nonisothermal SWRC model was derived to predict the water-retention capability of unsaturated rooted soils at different soil temperatures. An addition parameter, β , is needed to capture the temperature effect; it is soil-specific, but the investigation in this study showed that a single β can be used to predict the SWRC of the rooted soils at three different values of temperature reasonably well.
4. An HCF model was derived by integrating the newly developed RT-SWRC model into the existing **Mualem's (1976)** model, with an attempt to capture the temperature effects on HCF. The comparisons with the measurements showed that although the HCF model captured the trend qualitatively (i.e., lower relative hydraulic conductivity at higher temperature), quantitatively, the prediction was not satisfactory, leaving a rather large discrepancy at any temperature. This is primarily because the case of nonisothermal rooted soil did not fully obey the assumptions made in the **Mualem's (1976)** model about the randomness of soil pore size distribution. Due to the lack of data, it is currently unclear how the presence of roots might alter the soil pore size distribution and how this soil-root interaction might further be affected by temperature change.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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